

Course Code	: 2304CS301	Date	: 12-05-2026
Course Name	: Computer Network	Duration	: 150 Minutes
		Total Marks	: 70

Instructions:

1. Attempt all the questions.
2. Figures to the right indicates maximum marks.
3. Make suitable assumptions wherever necessary.

- Q.1 (A)** Compare LAN, MAN and WAN. **4**
- (B)** Write down the applications of computer network. **3**

OR

Write down the advantages and disadvantage of computer network.

- (C)** Explain star, ring and bus topologies with suitable figures. **7**

OR

Explain Delay, Loss and Throughput with suitable examples.

- Q.2 (A)** Compare STP and UTP. **4**
- (B)** Write a short note on Infrared waves. **3**

OR

Write a short note on Radio waves.

- (C)** Explain co-axial cable with a suitable diagram. **7**

OR

Draw a neat and clean diagram of a fiber-optic cable and explain each layer of it.

- Q.3 (A)** Compare circuit switching and packet switching. **4**
- (B)** Explain Half-duplex communication mode with a suitable example. **3**

OR

Explain Full-duplex communication mode with a suitable example.

- (C)** Explain the OSI reference model with a suitable figure. **7**

OR

Explain the TCP/IP model and also compare the OSI and TCP/IP models.

- Q.4 (A)** Write a short note on an Access Point. **4**
- (B)** Write a short note on a LAN Card. **3**

OR

Write a short note on a Modem.

- (C)** Explain Switch and Bridge networking devices and also compare both devices. **7**

OR

List the different types of servers and explain the Print server, Mail server and Web server with suitable figures.

- Q.5 (A)** Draw a neat and clean diagram of the IPv4 header format. **4**
- (B)** Identify the class of the following IP addresses: **3**
1. 10.0.0.1
 2. 220.45.67.89
 3. 239.255.1.1

OR

Identify the class of the following IP addresses:

1. 250.1.2.3
2. 150.10.20.30
3. 192.168.1.10

- (C)** Explain Classful Addressing of IP addresses with a proper example. **7**

OR

Explain the following terms:

1. Email
2. DNS
3. SMTP
4. FTP
5. WWW
6. HTTP
7. Web Hosting
